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An integral formula for topological entropy of C^∞ maps

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Abstract. In this paper we consider a smooth dynamical system f and give estimates of the growth rates of vector fields and differential forms in the L_p norm under the action of the dynamical system in terms of entropy, topological pressure and Lyapunov exponents. We prove a formula for the topological entropy

$$h_{\text{top}} = \lim_{n\to\infty} \frac{1}{n} \log \int \|Df_x^{n\,\wedge}\|\, dx,$$

where $Df_x^{n\,\wedge}$ is a mapping between full exterior algebras of the tangent spaces. An analogous formula is given for the topological pressure.

1. *Introduction*

Let X be a compact smooth manifold and let $f : X \hookleftarrow$ be a diffeomorphism of this manifold. The growth rate of the L_p norm of a differential form or a vector (multi-vector) field under iterations of the diffeomorphism f is of great importance in physics and also useful in the study of dynamical systems. By the growth rate we mean the limit $\lim_{n\to\infty}(1/n)\log\|f^n(\omega)\|_{L_p}$, where ω is some object and $\|\omega\|_{L_p} = (\int_X \|\omega\|^p\, dx)^{1/p}$ is the usual L_p norm. This limit may not exist so we have to deal with lim sup and lim inf instead.

As one of the consequences of the results we obtain the following formula for the topological entropy of a smooth mapping f:

$$h_{\text{top}} = \lim_{n\to\infty} \frac{1}{n} \log \int_X \|Df_x^{n\,\wedge}\|\, dx, \tag{1}$$

where $Df_x^{n\,\wedge}$ is a mapping between full exterior algebras of the tangent spaces, and we integrate with respect to the Lebesgue measure, which does not need to be invariant under f. However, if f possesses an absolute continuous invariant measure μ, it is interesting to compare this formula to the well-known Pesin formula of the metric entropy:

$$h_\mu = \lim_{n\to\infty} \frac{1}{n} \int_X \log \|Df_x^{n\,\wedge}\|\, d\mu.$$

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As far as we know this remarkable formula (1) cannot be found in the literature. This is somewhat surprising because essentially all of the ingredients that we used for its proof are already contained in the work of Yomdin [**15**] and others. In fact, this equality has some history. The first attempt to compare h_{top} and $\limsup_{n\to\infty}(1/n)\log\int\|Df_x^n\wedge\|\,dx$ was made in [**14**]. Later, Misiurewicz found for each positive integer k a 2-sphere diffeomorphism of class C^k where h_{top} is strictly smaller than the limit. Przytycki proved the inequality $h_{\text{top}}\le\limsup\ldots$ for $C^{1+\alpha}$ diffeomorphisms in [**11**] and Newhouse extended it to the case of non-invertible maps in [**8**]. For one-dimensional maps the equality has been proved in [**7**].

After we had finished writing this paper, we received a preprint [**5**] where some results are given which are in the same direction.

Although many ideas we use have already appeared in some form in the scattered literature, we decided to give complete proofs so that full knowledge of the literature is not necessary to read this paper.

This formula can be useful not only in physics (for example, in dynamo theory—for a quick introduction to the subject see [**2**]) but also in numerical applications, because it provides a superior method for numerical calculation of the topological entropy.

Now let us be more precise and define the set of objects we shall consider.

- Let Ω^k be the space of continuous k-dimensional differential forms on the manifold X. For $\omega\in\Omega^k$, $1\le p\le\infty$, we define the growth rate ν as follows:

$$\begin{aligned}
\overline{\nu}^p(\omega,f) &= \limsup_{n\to\infty}\frac{1}{n}\log\|f^{n*}\omega\|_{L_p}\\
\underline{\nu}^p(\omega,f) &= \liminf_{n\to\infty}\frac{1}{n}\log\|f^{n*}\omega\|_{L_p}\\
\overline{\nu}^{*p}_k(f) &= \sup_{\omega\in\Omega^k}\overline{\nu}^p(\omega,f)\\
\underline{\nu}^{*p}_k(f) &= \sup_{\omega\in\Omega^k}\underline{\nu}^p(\omega,f).
\end{aligned}$$

- Set $T_x{}^{\wedge k}X=\wedge_{i=1}^k T_xX$ to be the kth exterior algebra of the tangent space. For a continuous multi-vector field $V\in T^{\wedge k}X$ the growth rate is defined in a similar way:

$$\begin{aligned}
\overline{\nu}^p(V,f) &= \limsup_{n\to\infty}\frac{1}{n}\log\|f^n{}_*V\|_{L_p}\\
\underline{\nu}^p(V,f) &= \liminf_{n\to\infty}\frac{1}{n}\log\|f^n{}_*V\|_{L_p}\\
\overline{\nu}^p_{*k}(f) &= \sup_{V\in T^{\wedge k}X}\overline{\nu}^p(V,f)\\
\underline{\nu}^p_{*k}(f) &= \sup_{V\in T^{\wedge k}X}\underline{\nu}^p(V,f)\,\dagger\,.
\end{aligned}$$

  If a map f is not invertible, we cannot define an action of f on vector fields. However, the growth rate has a sense in this case as well. For example, if V is a

† f_* and f^* act as usual. For example, if V is a vector field and ω is a one-dimensional differential form, then $(f_*V)(x)=Df_{f^{-1}x}V(f^{-1}x)$ and $(f^*\omega)(x)=Df_x^*\omega(fx)$, where Df_x is the derivative of f at x.

vector field, then

$$\int_X \|f_* V\|^p \, dx = \int_X \|Df_x V(x)\|^p \det Df_x \, dx.$$

So, we shall not assume the invertibility of f any more.

- Let $A : \mathbb{R}^m \hookleftarrow$ be a linear map. $A^{\wedge k} : \mathbb{R}^{m \wedge k} \hookleftarrow$ denotes the induced map on the kth exterior algebra of $\mathbb{R}^m$. $A^{\wedge}$ is the induced map on the full exterior algebra. The norm of $A^{\wedge k}$ has a simple geometrical meaning. Let $\mathrm{Vol}_k(v_1, \ldots, v_k)$ be the k-dim volume of a parallelepiped spanned by the vectors $v_1, \ldots, v_k$, where $v_i \in \mathbb{R}^m$. Then

$$\begin{aligned} \|A^{\wedge k}\| &= \sup_{v_i \in R^m} \frac{\mathrm{Vol}_k(Av_1, \ldots, Av_k)}{\mathrm{Vol}_k(v_1, \ldots, v_k)} \\ \|A^{\wedge}\| &= \max_{1 \le i \le m} \|A^{\wedge i}\|. \end{aligned}$$

The *k-dim variation in the L_p norm* of the mapping f is

$$\mathrm{var}_k^p(f) = \| \|Df^{\wedge k}\| \|_{L_p}.$$

We shall see that the growth rate of the multi-dimensional variation is connected to the growth rates of differential forms and multi-vector fields.

It is easy to see that the growth rate does not depend on the norm in the linear space $T_x{}^{\wedge k} X$ (or $T_x^{*\wedge k} X$). The measure that we take integrals with respect to should be absolutely continuous with respect to the Lebesgue measure and is not required to be invariant under the mapping f, although the same questions can be posed for some invariant measure μ.

The major aim of this paper is to give estimates for the growth rates in terms of the entropy, the topological pressure and the Lyapunov exponents of the mapping. We recall their definition in §3.

One can compare the growth rate and the logarithm of the spectral radius of the map f as it acts on the space $L_p(\Omega^k)$ or $L_p(T^{\wedge k} X)$. Set

$$\begin{aligned} r^{*k}_p &= \lim_{n \to \infty} \frac{1}{n} \log \|f^{*n}\|_{L_p(\Omega^k)} \\ r^{k}_{*p} &= \lim_{n \to \infty} \frac{1}{n} \log \|f_*{}^n\|_{L_p(T^{\wedge k} X)}. \end{aligned}$$

It is easy to see that in this case we do not need to use lim inf and lim sup. Obviously, $\overline{v}^{*p}_k \le r^{*k}_p$ and $\overline{v}_{*k}^{\,p} \le r^k_{*p}$; however, these inequalities can be strict. The map $f : [0,1] \hookleftarrow$, $fx = 2x - x^2$ (see Figure 1) provides such an example. For this map $\overline{v}_{*1}^{\,1} = 0$, but $r^1_{*1} = \log Df_0 = \log 2$.

Certainly, all of the results of this paper can be applied to the case of flows or semi-flows. The formulations of the theorems should be changed in the obvious way.

2. *General properties*

In this section we consider general properties of the growth rate for an arbitrary C^1 map $f : X \hookleftarrow$. First, $\overline{v}(\cdot, f^n) = n\overline{v}(\cdot, f)$ and $\underline{v}(\cdot, f^n) = n\underline{v}(\cdot, f), n > 0$; this is obvious. Later on we shall suppress the dependence on f, if it does not lead to a misunderstanding.

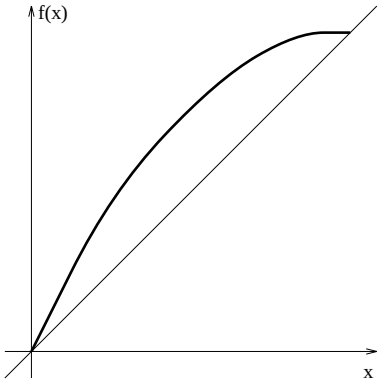


FIGURE 1. The map f.

THEOREM 2.1. *For generic† differential forms* $\omega \in \Omega^k$,

$$\overline{\nu}^p(\omega) = \overline{\nu}^{*p}_k \quad \textit{and} \quad \underline{\nu}^p(\omega) = \underline{\nu}^{*p}_k.$$

The same holds for multi-vector fields: for generic $V \in T^{\wedge k}X$

$$\overline{\nu}^p(V) = \overline{\nu}_{*k}^{\ p} \quad \textit{and} \quad \underline{\nu}^p(V) = \underline{\nu}_{*k}^{\ p}.$$

THEOREM 2.2.

$$\begin{aligned}
\overline{\nu}^{*p}_k &= \limsup_{n\to\infty} \frac{1}{np} \log \int_X \|Df_x^{n\wedge k}\|^p \, dx \\
\underline{\nu}^{*p}_k &= \liminf_{n\to\infty} \frac{1}{np} \log \int_X \|Df_x^{n\wedge k}\|^p \, dx \\
\overline{\nu}_{*k}^{\ p} &= \limsup_{n\to\infty} \frac{1}{np} \log \left| \int_X \|Df_x^{n\wedge k}\|^p \det Df_x^n \, dx \right| \\
\underline{\nu}_{*k}^{\ p} &= \liminf_{n\to\infty} \frac{1}{np} \log \left| \int_X \|Df_x^{n\wedge k}\|^p \det Df_x^n \, dx \right|.
\end{aligned}$$

In order not to overload the notation we shall prove these theorems only for vector fields and one-dimensional differential forms. It will be quite clear how to extend the proofs to higher dimensions.

First, we introduce the notion of generalized Lyapunov exponents and consider their properties. Fix p. Let $x \in X$ be a point of the manifold X and let $l \subset T_xX$ be a one-dimensional linear subspace of T_xX. The *generalized Lyapunov exponent* $\overline{\lambda}^p(x, l, f)$ is the infimum of the growth rates $\overline{\nu}^p(V, f)$ over all C^0 vector fields V such that at the point x the non-zero vector $V(x)$ lies in the space l:

$$\overline{\lambda}^p(x, l, f) = \inf\{\overline{\nu}^p(V, f) : V \in TX, V \in C^0, V(x) \in l, V(x) \neq 0\}.$$

LEMMA 2.1. *Let* $e_1, e_2, \ldots, e_{\dim X}$ *be a basis of* T_xX *satisfying the following properties:* $\overline{\lambda}^p(x, e_1) \geq \overline{\lambda}^p(x, e_2) \geq \cdots \geq \overline{\lambda}^p(x, e_{\dim X})$ *and if* $\overline{\lambda}^p(x, e_i) = \overline{\lambda}^p(x, e_{i+1}) = \cdots =$

† In this paper a 'generic' set means that the set is open and dense.

$\overline{\lambda}^p(x, e_m)$, *then* $\overline{\lambda}^p(x, e_i) = \overline{\lambda}^p(x, v)$, $v \in \langle e_k, \dots, e_m \rangle$. *If* $w \in T_x X$, $w = \sum a^i e_i$, *then* $\overline{\lambda}^p(x, w) = \overline{\lambda}^p(x, e_l)$, *where* k *is the index of the first non-zero coefficient* a_i: $a_i = 0$, $i < k$ *and* $a_k \neq 0$.

Proof. Fix x. Given small $\delta > 0$, choose $\dim X$ vector fields $\{V_i\}$ such that $V_i(x) = e_i$ and $\overline{\lambda}^p(x, e_i) + \delta > \overline{v}^p(V_i)$. In a small neighborhood U of the point x the vector fields $\{V_i\}$ form a basis for TU. Let W be a vector field, $W(y) = \sum a^i V_i(y)$, where $y \in X$. The following inequality is obvious: $\|f_*^n W\|_{L_p} = \|\sum a^i f_*^n V_i\|_{L_p} \leq \sum |a^i| \|f_*^n V_i\|_{L_p}$. So, $\overline{v}^p(W) \leq \overline{v}^p(V_l)$, hence, $\overline{\lambda}^p(x, w) < \overline{\lambda}^p(x, e_l) + \delta$, and $\overline{\lambda}^p(x, w) \leq \overline{\lambda}^p(x, e_l)$ because δ is arbitrarily small.

Now let us prove the opposite inequality $\overline{\lambda}^p(x, w) \geq \overline{\lambda}^p(x, e_l)$. Take an arbitrary vector field W, $W(x) = w$. Let m be the greatest index such that $\overline{\lambda}^p(x, e_m) = \overline{\lambda}^p(x, e_l)$. Consider the projection w' of the vector w onto the linear space spanned on the vectors $e_l, \dots, e_m$. Obviously, $w' = \sum_{i=l}^m a^i e_i$. Decompose the vector field W as

$$W(y) = W'(y) + \sum_{i=m+1}^{\dim X} a^i(y) V_i(y), \quad y \in X,$$

where $W'(x) = w', a^i(x) = a^i$. We know that $\overline{v}^p(W') \geq \overline{\lambda}^p(x, w) = \overline{\lambda}^p(x, e_l)$ and $\overline{v}^p(V_i) < \overline{\lambda}^p(x, e_l)$, if δ is small enough. The inequality

$$\|f_*^n W\|_{L_p} \geq \|f_*^n W'\|_{L_p} - C \sum_{i=m+1}^{\dim X} \|f_*^n V_i\|_{L_p},$$

where $C = \max_i \sup_y |a_i(y)|$, implies that $\overline{v}^p(W) \geq \overline{v}^p(W') \geq \overline{\lambda}^p(x, e_l)$. □

Such a basis $e_1, e_2, \dots, e_{\dim X}$ will be called *a nice basis.*

LEMMA 2.2. *For any point* $x \in X$ *there is a sequence of nested linear subspaces* $T_x X = E_1^p(x) \supset E_2^p(x) \supset \cdots \supset E_{r(x)}^p(x)$ *and numbers* $\overline{\lambda}_1^p(x) > \overline{\lambda}_2^p(x) > \cdots > \overline{\lambda}_{r(x)}^p(x)$ *such that if* $l \subset E_i^p(x) \setminus E_{i+1}^p(x)$, *then* $\overline{\lambda}^p(x, l) = \overline{\lambda}_i^p(x)$.

Proof. Take a basis in $T_x X$. This basis can be transformed into a nice basis in a finite number of steps. For such a basis the existence of the subspaces E_i is obvious. □

Certainly, the functions $\overline{\lambda}_i^p$, $\dim E_i^p$ and the spaces $E_i^p(x)$ are f-invariant.

LEMMA 2.3. *The function* $\overline{\lambda}_1^p$ *is upper semi-continuous.*

Proof. Let $x_1, x_2, \dots$ be a converging sequence of points of X, $x_i \to x_0$. Take a subsequence $\{x_{i_j}\}$ of $\{x_i\}$ such that $\limsup \overline{\lambda}_1^p(x_i) = \limsup \overline{\lambda}_1^p(x_{i_j})$, $\dim E_2^p(x_{i_j})$ is constant and the sequence $\{E_2^p(x_i)\}$ converges to $E \subset T_{x_0} X$. Relabel the subsequence $\{x_i\}$. If a vector w does not lie in the space E, then for any vector field W, $W(x_0) = w$, and any number N there is a point x_i such that $W(x_i) \notin E_2^p(x_i)$ and $i > N$. This implies $\overline{v}^p(W) \geq \overline{\lambda}_1^p(x_i)$, so $\overline{\lambda}_1^p(x_0) \geq \limsup \overline{\lambda}_1^p(x_i)$. □

One can easily construct an example where the function $\overline{\lambda}_1^p$ is not continuous.

LEMMA 2.4.

$$\overline{\nu}_{*1}^{\,p} = \max_{x\in X} \overline{\lambda}_1^p(x).$$

Proof. The maximum exists because of the upper semi-continuity of the function $\overline{\lambda}_1^p$. The inequality $\overline{\nu}_{*1}^{\,p} \geq \max_{x\in X} \overline{\lambda}_1^p(x)$ is obvious, so we have to prove the other inequality. Let W be a vector field. For any open set $U \subset X$ define the vector field W_U as follows: $W_U(x) = W(x)$ if $x \in U$ and $W_U(x) = 0$ if $x \notin U$. Take a point $x_0 \in X$ such that for any neighborhood U of the point x_0 we have $\overline{\nu}^p(W_U) = \overline{\nu}^p(W)$. If there is no such point, then for any point x there is a neighborhood U_x such that $\overline{\nu}^p(W_{U_x}) < \overline{\nu}^p(W)$. Since the manifold X is compact we can take a finite covering $\{U_x\}$ of X and obtain $\overline{\nu}^p(W) = \max \overline{\nu}^p(W|_{U_x}) < \overline{\nu}^p(W)$, which is a contradiction. Now, for given $\delta > 0$, take a nice basis at the point x_0 and vector fields V_i as in the proof of Lemma 2.1. These vector fields form a basis for TU, where U is a small enough neighborhood of the point x. Let $W(y) = \sum a^i(y)V_i(y)$, $y \in U$ and $C > |a^i(y)|$. Then $\|f_*^n W_U\|_{L_p} \leq C \sum_i \|f_*^n V_i\|_{L_p}$, and $\overline{\nu}^p(W) = \overline{\nu}^p(W_U) \leq \overline{\lambda}_1^p(x_0)$. □

The proof of Theorem 2.1 follows from Lemma 2.4. $\overline{\nu}^p(V) = \overline{\nu}_{*1}^{\,p}$ if and only if $V(x) \in E_1(x)$ where x is a point such that $\overline{\lambda}_1^p(x) = \overline{\nu}_{*1}^{\,p}$. This is a generic condition.

Let ω be a one-dimensional differential form and let $\|\omega(x)\| < C$. The inequality

$$\|f^{n*}\omega\|_{L_p} = \left(\int_X \|(Df_x^n)^*\omega(f^n(x))\|^p\,dx\right)^{1/p} \leq C\left(\int_X \|Df_x^n\|^p\,dx\right)^{1/p}$$

implies $\overline{\nu}^{*p}_1 \leq \limsup_{n\to\infty}(1/np)\log\int_X \|Df_x^{n\wedge k}\|^p\,dx$. In order to prove the inverse inequality take a point x_0 such that for any neighborhood U of x_0 we have $\limsup(1/np)\log\int_{f^{-n}U}\|Df_x^n\|^p = \overline{\nu}^{*p}_1$. The arguments used in the proof of Lemma 2.4 show the existence of such a point. Consider a chart of some neighborhood U around the point x_0 with coordinates $z_1, \ldots, z_{\dim X}$. Define a one-form ω, $\omega = dz_i$ in U and $\omega = 0$ outside of U. Let us estimate the ji-element of the matrix $Df^n_{f^{-n}x}$.

$$\int_{f^{-n}U} |(Df_z^n)_{ji}|^p\,dz \leq C_1 \int_{f^{-n}U} \|(Df_z^n)_i\|^p\,dz = C_1 \int_{f^{-n}U} \|f^{n*}\omega\|^p\,dz,$$

where $(Df_z^n)_i$ denotes the ith row of the matrix Df_z^n. We know that $\max_{ij}|A_{ij}| > C_2\|A\|$. Hence, $\overline{\nu}^{*p}_1 \geq \limsup_{n\to\infty}(1/np)\log\int_X \|Df_x^{n\wedge k}\|^p\,dx$. This proves the required equality in the one-dimensional case. In the multi-dimensional case one should use the following lemma instead of the inequality $\max_{ij}|A_{ij}| > C_2\|A\|$.

LEMMA 2.5. *Let A be an $n \times n$ matrix. Then there is a constant $C > 0$ depending on n such that*

$$C\|A^{\wedge k}\| \leq \max\{|m| : m \text{ is a minor of } A \text{ of } k\text{th order}\} \leq \|A^{\wedge k}\|.$$

This completes the proof of Theorem 2.2. The case of $\underline{\nu}$ can be treated analogously.

3. *Estimates of the growth rate*

3.1. *Statement of results.* First we recall some basic notions of ergodic theory. Let d be the metric on X. For $n = 0, 1, \ldots$ define a new metric $d_{f,n}$ on X by

$$d_{f,n}(x, y) = \max_{i=0,1,\ldots,n} d(f^i x, f^i y).$$

We shall need two standard definitions of the topological entropy. A set is called (n, ε)-*separated* if ε-balls in the $d_{f,n}$-metric centered at the points of the set do not overlap each other. Denote by $r(n, \varepsilon, f)$ the cardinality of the maximal (n, ε)-separated set. Then the *topological entropy* can be defined by

$$h_{\text{top}}(f) = \lim_{\varepsilon \to 0} \limsup_{n \to \infty} \frac{1}{n} \log r(n, \varepsilon, f).$$

A set is called (n, ε)-*spanning* if ε-balls in the $d_{f,n}$-metric centered at the points of the set cover the manifold X. Denote by $s(n, \varepsilon, f)$ the cardinality of the minimal (n, ε)-spanning set. Then the topological entropy is

$$h_{\text{top}}(f) = \lim_{\varepsilon \to 0} \limsup_{n \to \infty} \frac{1}{n} \log s(n, \varepsilon, f).$$

One can easily show that these two definitions give the same number.

There are several definitions of the metric entropy. We shall use one due to Katok [**3**]. Let μ be an invariant ergodic Borel probability measure. For $\varepsilon > 0$, $1 > \sigma > 0$, let us denote by $s(n, \varepsilon, \sigma, \mu, f)$ the minimal number of ε-balls in the d_n metric which cover a set of measure greater than or equal to σ. Then, for any $1 > \sigma > 0$, the *metric entropy* is

$$h_\mu(f) = \lim_{\varepsilon \to 0} \liminf_{n \to \infty} \frac{1}{n} s(n, \varepsilon, \sigma, \mu, f) = \lim_{\varepsilon \to 0} \limsup_{n \to \infty} \frac{1}{n} s(n, \varepsilon, \sigma, \mu, f).$$

The well known variational principal for the entropy says that the topological entropy is a supremum of the metric entropies over all invariant measures:

$$h_{\text{top}}(f) = \sup_{\mu \in \mathcal{M}(f)} h_\mu.$$

The point $x \in X$ is said to be *regular* if there exist numbers $\lambda_1(x) > \cdots > \lambda_{r(x)}(x)$ and a decomposition of the tangent space at x into $T_x X = E_1(x) \oplus \cdots \oplus E_{r(x)}(x)$ such that for every tangent vector $v \in E_i(x)$, $v \neq 0$,

$$\begin{aligned} \lim_{n \to \infty} \frac{1}{n} \log \| D f_x^n v \| &= \lambda_i(x) \\ \lim_{n \to \infty} \frac{1}{n} \log \det D f_x^n &= \sum_{i=1}^{r(x)} \lambda_i(x) \dim E_i(x). \end{aligned}$$

By a theorem of Oseledec, the set Γ of regular points has full measure for any measure $\mu \in \mathcal{M}(f)$. The numbers $\lambda_i(x)$ are called the *Lyapunov exponents* of f at x; $\dim E_i(x)$ is called the multiplicity of $\lambda_i(x)$. The number of positive Lyapunov exponents will be denoted by $u(x)$. The functions r, λ_i, u, $\dim E_i$ are measurable and invariant along orbits, and so are constant almost everywhere if μ is ergodic.

Sometimes we shall use an unusual indexing of the Lyapunov exponents. We will allow equal exponents and write a prime in this case. So let $\mathcal{M}_e$ denotes all f-invariant ergodic Borel probability measures, $\mu \in \mathcal{M}_e$ and let $\lambda'_1(\mu) \geq \lambda'_2(\mu) \geq \cdots \geq \lambda'_{\dim X}(\mu)$ be the Lyapunov exponents. In this sequence each Lyapunov exponent appears as many times as its multiplicity. Let $u'(\mu)$ denote a number of the positive Lyapunov exponents (with the multiplicities) and let $\lambda^+(\mu)$ ($\lambda^-(\mu)$) be a sum of the positive (negative) Lyapunov exponents.

$$\lambda^+(\mu) = \sum_{i=1}^{u'(\mu)} \lambda'_i(\mu), \quad \lambda^-(\mu) = \sum_{i=u'(\mu)+1}^{\dim X} \lambda'_i(\mu).$$

The following theorem gives the estimates of the growth rates from below.

THEOREM 3.1. *Let $f : X \hookleftarrow$ be a C^2 diffeomorphism. Then*

$$\begin{aligned} \underline{\nu}^{*p}_k &\geq \frac{1}{p} \sup_{\mu\in\mathcal{M}_e} \left(h_\mu - \lambda^+(\mu) + p \sum_{i=1}^{k} \lambda'_i(\mu) \right) \\ \underline{\nu}_{*k}^{\ p} &\geq \frac{1}{p} \sup_{\mu\in\mathcal{M}_e} \left(h_\mu + \lambda^-(\mu) + p \sum_{i=1}^{k} \lambda'_i(\mu) \right). \end{aligned}$$

In the important case of the L_1 norm we obtain as a consequence that there exists k such that $\underline{\nu}^{*1}_k \geq h_{\text{top}}$. As we shall see later, if the map f is infinitely differentiable, then $\underline{\nu}^{*1}_k = h_{\text{top}}$.

In the case of an Axiom A C^2 diffeomorphism f satisfying the transversality condition the inequalities in Theorem 3.1 are, in fact, equalities.

The mapping f is called the *Axiom A* diffeomorphism if its non-wandering set is hyperbolic and equal to the closure of the set of periodic points of f. The diffeomorphism f satisfies the *transversality condition* if the stable and unstable manifolds of the non-wandering set intersect transversally.

THEOREM 3.2. *If the Axiom A C^2 diffeomorphism f satisfies the transversality condition, then*

$$\begin{aligned} \underline{\nu}^{*p}_k &= \overline{\nu}^{*p}_k = \frac{1}{p} \sup_{\mu\in\mathcal{M}_e} \left(h_\mu - \lambda^+(\mu) + p \sum_{i=1}^{k} \lambda'_i(\mu) \right) \\ \underline{\nu}_{*k}^{\ p} &= \overline{\nu}_{*k}^{\ p} = \frac{1}{p} \sup_{\mu\in\mathcal{M}_e} \left(h_\mu + \lambda^-(\mu) + p \sum_{i=1}^{k} \lambda'_i(\mu) \right). \end{aligned}$$

If $\|Df_x\| < N$ for $x \in X$, we can estimate the growth rate as

$$\overline{\nu}^{*p}_k \leq k \log N, \quad \overline{\nu}_{*k}^{\ p} \leq k \log N.$$

However, we can bound the growth rates from above more precisely. For this purpose we shall need a notion of the topological pressure which generalizes the topological entropy in the following way. Let $\gamma : X \to \mathbb{R}$ be a C^0 function and let $\bar{\gamma}_n(x)$ be an average of the function γ along an orbit of length n, $\bar{\gamma}_n(x) = (1/n) \sum_{i=0}^{n-1} \gamma(f^i(x))$.

Set

$$\begin{aligned} r(n, \varepsilon, \gamma, f) &= \sup\left\{\sum_{x\in E} e^{\bar{\gamma}_n(x)} : E \text{ is } (n,\varepsilon)\text{-separated}\right\}, \\ r(\varepsilon, \gamma, f) &= \limsup_{n\to\infty} \frac{1}{n} \log r(n, \varepsilon, \gamma, f). \end{aligned}$$

The *topological pressure* $P_f(\gamma)$ is defined to be

$$P_f(\gamma) = \lim_{\varepsilon\to 0} r(\varepsilon, \gamma, f).$$

Obviously, $P(0) = h_{\text{top}}$.

The variational principle extends to the case of the topological pressure as

$$P_f(\gamma) = \sup_{\mu\in\mathcal{M}(f)} \left(h_\mu + \int_X \gamma \, d\mu\right).$$

This supremum can be taken over the set of all ergodic invariant measures as well.

THEOREM 3.3. *Let $f : X \hookleftarrow$ be a C^∞ map. Then*

$$\begin{aligned} \overline{v}^{*p}_k &\le \frac{1}{p} P(\log(\|Df_x{}^\wedge\|^{p-1})) \\ \overline{v}_{*k}^{\,p} &\le \frac{1}{p} P(\log(\|Df_x{}^\wedge\|^{p-1} |\det Df_x|)). \end{aligned}$$

The proof of this theorem will be given at the end of §4.

CONSEQUENCE 3.4. *Let $f \in C^\infty$, then*

$$\overline{v}^{*1}_k \le h_{\text{top}}. \tag{2}$$

There exist k such that (2) is an equality.

CONSEQUENCE 3.5. *Let $f \in C^\infty$, then*

$$h_{\text{top}} = \lim_{n\to\infty} \frac{1}{n} \log \int_X \|Df_x^{n\,\wedge}\| \, dx.$$

The requirement that f be C^∞ is essential. For any k there exists a map f such that Theorem 3.3 and Consequence 3.4 are wrong.

Example. Let $f : I \hookleftarrow$ be an odd map of an interval $I = [-1, 1]$ to itself such that $f(x) = 2x$ if $x \in [0, \frac{1}{4}]$, $f(x) = (x-1)^k \sin(\pi/(x-1))$ if $x \in [\frac{3}{4}, 1]$. Define f on $[\frac{1}{4}, \frac{3}{4}]$ in such a way that f is C^{k-3} and has only two fixed sinks (see Figure 2). Obviously, the non-wandering set of f is 3 fixed points, so the topological entropy of this map is equal to 0. Now let us estimate the growth rate of the variation of f.

Let $\text{var}(f, [a, b])$ denote the variation of f on the segment $[a, b]$. Then

$$\text{var}\left(x^k \sin\frac{\pi}{x}, [0, \varepsilon]\right) \sim 2 \sum_{i>\varepsilon^{-1}} i^{-k} \sim 2 \int_{\varepsilon^{-1}}^{\infty} x^{-k} \, dx = \frac{2}{k-1} \varepsilon^{k-1}.$$

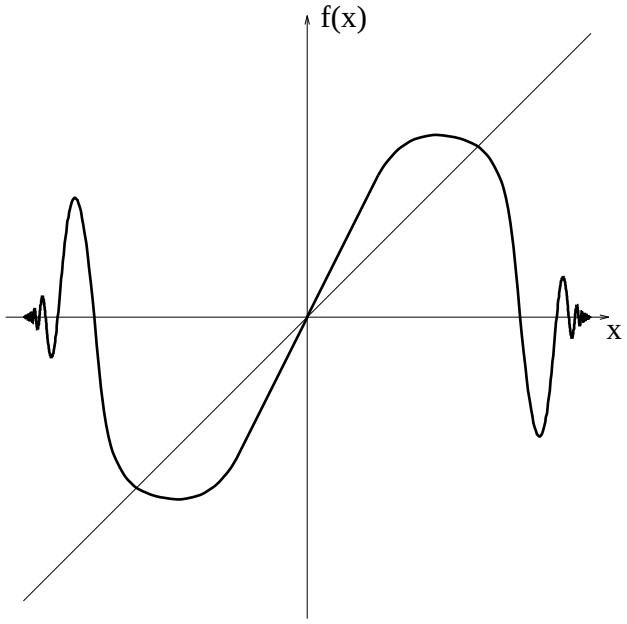


FIGURE 2. The map f.

If $|f(x)| < 2^{-(n+1)}$ then $f^n(x) = 2^n f(x)$, hence

$$\operatorname{var}(f^n, I) \geq \operatorname{var}\left(2^n x^k \sin\frac{\pi}{x}, [0, 2^{-(n+1)/k}]\right) \sim \frac{1}{k-1} 2^{(n+1)/k},$$

and

$$\liminf_{n\to\infty} \frac{1}{n} \log \operatorname{var}(f^n, I) \geq \frac{1}{k} \log 2.$$

3.2. *Lyapunov charts and some estimates.* In this subsection we apply some results from Pesin theory to the case of non-invertible maps. See Appendix A.1 for more details. As usual, d denotes the Riemannian metric on the manifold X. Let $\hat{X} = \{\hat{x} = (x_0, x_1, \ldots) : x_i \in X \text{ and } fx_{i+1} = x_i \text{ for } i \geq 0\}$ and let $\hat{f} : \hat{X} \hookleftarrow$ be defined by $\hat{f}((x_0, x_1, \ldots)) = (fx_0, x_0, x_1, \ldots)$. Give $\hat{X}$ the metric $d(\hat{x}, \hat{y}) = \sum_{i=0}^{\infty} 2^{-i} d(x_i, y_i)$. In this metric the map $\hat{f}$ is a homeomorphism and is called the *inverse limit* of the map f.

Let $\pi : \hat{X} \to X$ be the natural projection, $\pi(\hat{x}) = x_0$. For any $\mu \in \mathcal{M}(f)$ there exists a unique measure $\hat{\mu} \in \mathcal{M}(\hat{f})$ such that $\pi_* \hat{\mu} = \mu$ [**12**].

We shall define in a non-autonomous way a change of coordinates in some neighborhood of each regular point. The size of the neighborhood, the local charts and the estimates will vary with x.

Fix an ergodic measure $\mu \in \mathcal{M}_e(f)$. There are numbers $\lambda_1 > \cdots > \lambda_r$ and a set of full measure Γ such that for any $x \in \Gamma$ there is a splitting of the tangent space at x into $T_x X = E_1(x) \oplus \cdots \oplus E_{r(x)}(x)$ such that for every tangent vector $v \in E_i(x)$, $v \neq 0$,

$$\lim_{n\to+\infty} \frac{1}{n} \log \|Df_x^n v\| = \lambda_i(x).$$

The exponent λ_r may be equal to $-\infty$.

Let $\mathbb{R}^{\dim X} = \mathbb{R}^{\dim E_1} \times \cdots \times \mathbb{R}^{\dim E_r}$ and for $y \in \mathbb{R}^{\dim X}$, let $(y_1, \ldots, y_r)$ be its coordinates with respect to this splitting. Define

$$\begin{aligned} |y| &= \max_{1\le i\le r} |y_i|_{\dim E_i} \\ |y|_u &= \max_{1\le i\le u} |y_i|_{\dim E_i} \\ |y|_s &= \max_{u<i\le r} |y_i|_{\dim E_i}, \end{aligned}$$

where $|\cdot|_{\dim E_i}$ is the Euclidean norm on $\mathbb{R}^{\dim E_i}$. The closed disk in $\mathbb{R}^{\dim X}$ of radius ρ is denoted by $R(\rho)$:

$$R(\rho) = \{y \in \mathbb{R}^{\dim X} : |y| \le \rho\}.$$

Given $\varepsilon < (1/100)\min_{a\ne b\in S}|a-b|$ where $S = \{0\}\cup\{\lambda_1, \ldots, \lambda_r\}$, there is a set $\hat{\Gamma} \subset \hat{X}$ of full measure ($\hat{\mu}\hat{\Gamma} = 1$), $\pi\hat{\Gamma} \subset \Gamma$, and a measurable function $\rho : \hat{\Gamma} \to (0, 1]$ such that

$$\rho(\hat{f}^{\pm 1}\hat{x}) \ge e^{-\varepsilon}\rho(\hat{x}). \tag{3}$$

Then there is an embedding $\Phi_{\hat{x}} : R(\rho(\hat{x})) \to X$ with the following properties:

- $\Phi_{\hat{x}}(0) = \pi(\hat{x})$, and $D\Phi_{\hat{x}}(0)$ takes $\mathbb{R}^{\dim E_i}$ to E_i for $i = 1, \ldots, r$.
- $\exp^{-1}_{\pi\hat{x}} \circ \Phi_{\hat{x}}$ coincides with $D\Phi_{\hat{x}}(0)$ on $R(\rho(\hat{x}))$. (This means that $\Phi_{\hat{x}}$ is, in some sense, a linear map.)
- Let $\tilde{f}_{\hat{x}} = \Phi^{-1}_{\hat{f}\hat{x}} \circ f \circ \Phi_{\hat{x}}$ be the connecting map between the chart at $\pi(\hat{x})$ and the chart at $\pi(\hat{f}\hat{x})$, defined wherever it makes sense. Then for all $v \in \mathbb{R}^{\dim E_i}$,

  $$e^{\lambda_i-\varepsilon}|v| \le |D\tilde{f}_{\hat{x}}(0)\, v| \le e^{\lambda_i+\varepsilon}|v|.$$

  If $\lambda_r = -\infty$, then for all $v \in \mathbb{R}^{\dim E_r}$,

  $$|D\tilde{f}_{\hat{x}}(0)\, v| \le e^{-\varepsilon^{-1}}|v|.$$

- If $\mathrm{Lip}(g)$ denotes the Lipschitz constant of the function g, then

  $$\begin{aligned} \mathrm{Lip}(\tilde{f}_{\hat{x}} - D\tilde{f}_{\hat{x}}(0)) &\le \varepsilon \\ \mathrm{Lip}(D\tilde{f}_{\hat{x}}) &\le \rho(\hat{x})^{-1}. \end{aligned}$$

- For all $z, z' \in R(\rho(\hat{x}))$,

  $$\rho(\hat{x})|z - z'| \le d(\Phi_{\hat{x}}z, \Phi_{\hat{x}}z') \le K_r|z - z'|$$

  for some universal constant K_r.
- For $y \in R(\rho(\hat{x}))$,

  $$\begin{aligned} \|D\Phi_{\hat{x}}(y)\| &\le K_r \\ \|D\Phi^{-1}_{\hat{x}}(y)\| &\le \rho(\hat{x})^{-1}. \end{aligned} \tag{4}$$

The following estimate is used for the connecting maps between the charts.

LEMMA 3.1. *Let $\mathbb{R}^{\dim X} = \mathbb{R}^{\dim E_1} \times \cdots \times \mathbb{R}^{\dim E_r}$ and let $\lambda_1, \ldots, \lambda_r$ be some numbers. Then there are positive constants ε_0 and C_1 depending on $\lambda_1, \ldots, \lambda_r$ such that for any*

$\varepsilon < \varepsilon_0$ if $A_i : \mathbb{R}^{\dim X} \hookleftarrow$, $i = 1, 2, \ldots$, are linear such that $A_i : \mathbb{R}^{\dim E_j} \to \mathbb{R}^{\dim E_j}$ and for all $v \in \mathbb{R}^{\dim E_i}$,

$$e^{\lambda_i - \varepsilon}|v| \le |A_i v| \le e^{\lambda_i + \varepsilon}|v|$$

and if $B_i : \mathbb{R}^{\dim X} \hookleftarrow$, $i = 1, 2, \ldots$, are linear such that $\|B_i - A_i\| < \varepsilon$, then

$$e^{-C_1\varepsilon n} e^{n\sum_{i=1}^{k}\lambda'_i} < \|(B_n \cdots B_1)^{\wedge k}\| < e^{C_1\varepsilon n} e^{n\sum_{i=1}^{k}\lambda'_i}.$$

For the proof of this lemma see Appendix A.2.

3.3. *Estimate from below (proof of Theorem 3.1).* Fix an invariant ergodic measure μ. We shall assume that $\lambda_r \neq -\infty$. This is not a strong restriction.

Let $B_x^n(\varepsilon)$ denote a ball of radius ε at the point x in metric d_n and $\tilde{B}_{\hat{x}}^n(\varepsilon) = \{y \in R(\rho(\hat{x})) : |\tilde{f}_{\hat{x}}^i(y)| < \varepsilon K_r^{-1} e^{-i\varepsilon}\rho(\hat{x}),\ i = 0, \ldots, n\}$. Since $\|D\Phi_{\hat{x}}(y)\| \le K_r$, we have $\Phi_{\hat{x}}\tilde{B}_{\hat{x}}^n(\varepsilon) \subset B_{\pi(\hat{x})}^n(\varepsilon)$. For $y \in \tilde{B}_{\hat{x}}^n(\varepsilon)$ a value of $\tilde{f}_{\hat{x}}^n(y)$ is defined and we have $\tilde{f}_{\hat{x}}^n(y) = \Phi_{\hat{f}^n(\hat{x})}^{-1} \circ f^n \circ \Phi_{\hat{x}}(y)$. Hence,

$$\begin{aligned}
\|D\tilde{f}_{\hat{x}}^n(y)^{\wedge k}\| &\le \|D\Phi_{\hat{x}}(y)\|^k \|D\Phi^{-1}_{\hat{f}^n(\hat{x})}(f^n(\Phi_{\hat{x}}(y)))\|^k \|Df^n(\Phi_{\hat{x}}(y))^{\wedge k}\| \\
&\le K_r^k \rho(\hat{f}^n\hat{x})^{-k} \|Df^n(\Phi_{\hat{x}}(y))^{\wedge k}\| \\
&\le K_r^k \rho(\hat{x})^{-k} e^{-\varepsilon k n} \|Df^n(\Phi_{\hat{x}}(y))^{\wedge k}\|,
\end{aligned}$$

because $\|(AB)^{\wedge k}\| \le \|A^{\wedge k}\|\,\|B^{\wedge k}\|$ and $\|A^{\wedge k}\| \le \|A\|^k$, if A, B are matrices. We also used inequalities (3) and (4).

So, if $x = \pi(\hat{x})$,

$$\begin{aligned}
\int_{B_x^n(\varepsilon)} \|Df^n(z)^{\wedge k}\|^p\, dz &\ge K_r^{-kp}\rho(\hat{x})^{kp} e^{\varepsilon kpn} \int_{B_x^n(\varepsilon)} \|D\tilde{f}_{\hat{x}}^n(\Phi_{\hat{x}}^{-1}(z))^{\wedge k}\|^p\, dz \\
&\ge K_r^{-kp}\rho(\hat{x})^{kp} e^{\varepsilon kpn} \int_{\tilde{B}_{\hat{x}}^n(\varepsilon)} \|D\tilde{f}_{\hat{x}}^n(y)^{\wedge k}\|^p \det D\Phi_{\hat{x}}(y)\, dy \\
&\ge K_r^{-kp}\rho(\hat{x})^{\dim X + kp} e^{\varepsilon kpn} \int_{\tilde{B}_{\hat{x}}^n(\varepsilon)} \|D\tilde{f}_{\hat{x}}^n(y)^{\wedge k}\|^p\, dy. \qquad (5)
\end{aligned}$$

We used the fact that for an $m \times m$ matrix A, $|\det A| \ge \|A^{-1}\|^{-m}$ and inequality (4).

Now we are going to estimate the volume of $\tilde{B}_{\hat{x}}^n(\varepsilon)$.

LEMMA 3.2. *There exist positive constants C_2 and C_3 such that*

$$\mathrm{Vol}(\tilde{B}_{\hat{x}}^n(\varepsilon)) = \int_{\tilde{B}_{\hat{x}}^n(\varepsilon)} dy > C_3\rho(\hat{x})^{\dim X}\varepsilon^{\dim X} e^{-C_2\varepsilon n} e^{-\lambda^+ n}.$$

Proof. $\tilde{f}_{\hat{x}}$ is a Lipschitz function, so if $|y|_u \le |y|_s$, then $|\tilde{f}_{\hat{x}}y|_s < |D\tilde{f}_{\hat{x}}y|_s + \varepsilon|y| \le |y|_s(e^{\varepsilon} + \varepsilon) < e^{2\varepsilon}|y|_s$. On the other hand, if $|y|_u \ge |y|_s$, then $|\tilde{f}_{\hat{x}}y|_u > |D\tilde{f}_{\hat{x}}y|_u - \varepsilon|y| > |y|_u(e^{\lambda_u - \varepsilon} - \varepsilon) > |y|_u$, for small enough ε. This means that the unstable cones $|y|_u \ge |y|_s$ are invariant.

Let

$$y^s = (\underbrace{0, \ldots, 0}_{u'}, y_{u'+1}, \ldots, y_{\dim X}) \quad \text{and} \quad |y^s| < \varepsilon K_r^{-1} e^{-3\varepsilon n}\rho(\hat{x}).$$

Consider a u'-dimensional plane P_{y^s} parallel to $\mathbb{R}^{\dim E_1} \times \cdots \times \mathbb{R}^{\dim E_u}$ and $y^s \in P_{y^s}$. The discussion above shows that if $|y|_s$ is small, then the point y can escape from the $\varepsilon K_r^{-1}\rho(\hat{f}^n\hat{x})$-ball only through the unstable directions. More precisely, if $y \in \partial P_{y^s} \cap \tilde{B}^n_{\hat{x}}(\varepsilon)$, then $|\tilde{f}^n_{\hat{x}} y|_u = \varepsilon K_r^{-1} e^{-\varepsilon n}\rho(\hat{x})$. This implies that

$$\begin{aligned}
&\int_{P_{y^s}\cap\tilde{B}^n_{\hat{x}}(\varepsilon)} \det(D\tilde{f}^n_{\hat{x}}(y))_{u'\times u'} dy_1 \cdots dy_{u'} \\
&\quad = \quad u'\text{-dim volume of the set } \{|y|_u = \varepsilon K_r^{-1} e^{-\varepsilon n}\rho(\hat{x})\} \\
&\quad = \quad C_4 \varepsilon^{u'} K_r^{-u'} e^{-\varepsilon u' n}\rho(\hat{x})^{u'},
\end{aligned}$$

where $\det(D\tilde{f}^n_{\hat{x}}(y))_{u'\times u'}$ is the $u' \times u'$ minor given by the determinant of the matrix obtained by the intersection of the first u' rows and the first u' columns of the matrix $D\tilde{f}^n_{\hat{x}}(y)$.

So,

$$\begin{aligned}
&\int_{\tilde{B}^n_{\hat{x}}(\varepsilon)} \|D\tilde{f}^n_{\hat{x}}(y)^{\wedge u'}\|\, dy \\
&\quad \geq \quad \int_{\tilde{B}^n_{\hat{x}}(\varepsilon)} \det(D\tilde{f}^n_{\hat{x}}(y))_{u'\times u'}\, dy \\
&\quad \geq \quad \int_{|y^s|<\varepsilon K_r^{-1}e^{-3\varepsilon n}\rho(\hat{x})} \left(\int_{P_{y^s}\cap\tilde{B}^n_{\hat{x}}(\varepsilon)} \det(D\tilde{f}^n_{\hat{x}}(y))_{u'\times u'}\, dy_1 \cdots dy_{u'} \right) dy_{u'+1} \cdots dy_{\dim X} \\
&\quad > \quad C_4 \varepsilon^{u'} K_r^{-u'} e^{-\varepsilon u' n}\rho(\hat{x})^{u'} (\varepsilon K_r^{-1} e^{-3\varepsilon n}\rho(\hat{x}))^{\dim X - u'} \\
&\quad > \quad C_4 K_r^{-\dim X} \varepsilon^{\dim X} e^{-3\dim X \varepsilon n}\rho(\hat{x})^{\dim X}. \qquad (6)
\end{aligned}$$

The same quantity can be estimated from above as

$$\begin{aligned}
\int_{\tilde{B}^n_{\hat{x}}(\varepsilon)} \|D\tilde{f}^n_{\hat{x}}(y)^{\wedge u'}\|\, dy \quad &< \quad \mathrm{Vol}(\tilde{B}^n_{\hat{x}}(\varepsilon)) \max_{y\in\tilde{B}^n_{\hat{x}}(\varepsilon)} \|D\tilde{f}^n_{\hat{x}}(y)^{\wedge u'}\| \\
&< \quad e^{C_1\varepsilon n} e^{n\sum_{i=1}^{u'}\lambda'_i} \mathrm{Vol}(\tilde{B}^n_{\hat{x}}(\varepsilon)). \qquad (7)
\end{aligned}$$

We used Lemma 3.1. Combining (6) and (7) we obtain

$$\mathrm{Vol}(\tilde{B}^n_{\hat{x}}(\varepsilon)) > C_4 K_r^{-\dim X} \varepsilon^{\dim X} \rho(\hat{x})^{\dim X} e^{-(3\dim X + C_1)\varepsilon n} e^{n\sum_{i=1}^{u'}\lambda'_i}. \qquad \square$$

Combining Lemma 3.1, Lemma 3.2 and inequality (5) we obtain

$$\int_{B^n_x(\varepsilon)} \|Df^n(z)^{\wedge k}\|^p\, dz > C_5 \varepsilon^{C_6} \rho(\hat{x})^{C_6} e^{-C_7\varepsilon n} e^{n(\lambda^+ + p\sum_{i=1}^{k}\lambda'_i)}.$$

Let $\hat{\Lambda}(q) = \{\hat{x} \in \hat{X} : \rho(\hat{x}) > q\}$ and $\Lambda(q) = \pi\hat{\Lambda}(q)$. Obviously, $\mu\Lambda(q) \to 1, q \to 0$.

$$\begin{aligned}
\int_X \|Df^n(x)^{\wedge k}\|^p\, dx \quad &> \quad \sum_{x\in(\varepsilon,n,\Lambda(q))\text{-separated}} \int_{B^n_x(\varepsilon)} \|Df^n(z)^{\wedge k}\|^p\, dz \\
&> \quad s(n, 2\varepsilon, \mu\Lambda(q), \mu) C_5 \varepsilon^{C_6} q^{C_6} e^{-C_7\varepsilon n} e^{n(\lambda^+ + p\sum_{i=1}^{k}\lambda'_i)},
\end{aligned}$$

because any minimal $(n, 2\varepsilon)$-spanning set is (n, ε)-separated. Thus,

$$\begin{aligned} \underline{\nu}^{*p}_k &= \liminf_{n\to\infty} \frac{1}{n} \log \left(\int_X \|Df^n(x)^{\wedge k}\|^p \, dx \right)^{1/p} \\ &\geq \frac{1}{p} \left(\liminf_{n\to\infty} \frac{1}{n} s(n, 2\varepsilon, \mu\Lambda(q)) + \lambda^+ + p \sum_{i=1}^{k} \lambda'_i - C_7\varepsilon \right). \end{aligned}$$

Letting ε tend to 0 we obtain

$$\underline{\nu}^{*p}_k \geq \frac{1}{p} \sup_{\mu\in\mathcal{M}_e} \left(h_\mu - \lambda^+(\mu) + p \sum_{i=1}^{k} \lambda'_i(\mu) \right).$$

The estimate for the growth rate of multi-vector field can be proved analogously.

3.4. *The Axiom A case.* For each point $x \in \Omega(f)$ we can define a function $\phi_k(x)$ which is equal to the Jacobian of the restriction of Df_x on the sum of the first k spaces $E_i(x)$ counting with their multiplicities, and $\phi(x)$ is the Jacobian of Df_x restricted onto the unstable direction. These functions can be extended to the whole manifold X in such a way that ϕ_x and ϕ are Hölder continuous and there is constant C such that

$$\|Df^{n\,\wedge k}_x\| \leq C \prod_{i=0}^{n-1} \phi_k(f^i x).$$

Using the variational principal it is easy to show that

$$\sup_{\mu\in\mathcal{M}_e} \left(h_\mu - \lambda^+(\mu) + p \sum_{i=1}^{k} \lambda'_i(\mu) \right) = P(\log(\phi_k^p \phi^{-1})).$$

So, we have only to prove the inequality

$$\overline{\nu}^{*p}_k \leq \frac{1}{p} P(\log(\phi_k^p \phi^{-1})),$$

because Theorem 3.1 gives us the reversed one.

The following lemma first appeared in [**1**] and in the present form in [**4**].

LEMMA 3.3. (Volume lemma.) *Suppose that f is a C^2 Axiom A diffeomorphism satisfying the transversality condition, and that $\varepsilon > 0$ is a small positive number. Then there exists a constant $C_\varepsilon > 0$ depending on f and ϕ such that for any $x \in X$ and n,*

$$C_\varepsilon^{-1} \leq \mathrm{Vol}(B^n_x(\varepsilon)) \prod_{i=0}^{n} \phi(f^i x) \leq C_\varepsilon.$$

Here $B^n_x(\varepsilon)$ denotes a ball of radius ε in metric d_n at the point x, and Vol is the Riemannian volume.

If the set E_1 is $(n, \varepsilon/2)$-separated but not (n, ε)-spanning, we can add to this set several points in such a way that the new set E_2 remains $(n, \varepsilon/2)$-separated and becomes

(n, ε)-spanning. Certainly, $\sum_{x\in E_1} e^{n\bar{\gamma}_n(x)} \leq \sum_{x\in E_2} e^{n\bar{\gamma}_n(x)}$. Now we can estimate the growth rate as follows:

$$\begin{aligned}
\int_X \|Df_x^{n\,\wedge k}\|^p\,dx &\leq \sum_{\substack{x\in(n,\varepsilon/2)\text{-separated}\\ \text{and } (n,\varepsilon)\text{-spanning set}}} \int_{B_x^n(\varepsilon)} \|Df_y^{n\,\wedge k}\|^p\,dy \\
&\leq \max_{E-\ \substack{(n,\varepsilon/2)\text{-separated and}\\ (n,\varepsilon)\text{-spanning set}}} \sum_{x\in E} \mathrm{Vol}(B_x^n(\varepsilon))\|Df_x^{n\,\wedge k}\|^p \\
&\leq C_\varepsilon C^p \max_{E-\ \substack{(n,\varepsilon/2)\text{-separated and}\\ (n,\varepsilon)\text{-spanning set}}} \sum_{x\in E}\prod_{i=0}^{n} \phi_k(f^i x)^p \phi(f^i x)^{-1} \\
&\leq C_\varepsilon C^p r(n, \varepsilon/2, \log(\phi_k(f^i x)^p \phi(f^i x)^{-1}), f).
\end{aligned}$$

By taking $(1/n)\log$ and letting n tend to the infinity and ε to 0 we complete the proof.

4. *Topological pressure*

The topological pressure plays an important role in dynamical systems, so we give a formula for it.

THEOREM 4.1. *If f is a C^∞ map, then*

$$P_f(\gamma) = \lim_{n\to\infty} \frac{1}{n} \log \int_X e^{n\bar{\gamma}_n(x)} \|Df_x^{n\,\wedge}\|\,dx.$$

Proof.

Part 1. First, using the variational principle we prove that

$$P_f(\gamma) \leq \lim_{n\to\infty} \frac{1}{n} \log \int_X e^{n\bar{\gamma}_n(x)} \|Df_x^{n\,\wedge}\|\,dx.$$

Essentially, we shall follow §3.3.

For $\mu \in \mathcal{M}_e$, the functions $\bar{\gamma}_n$ converge μ-a.e. to $\int_X \gamma\,d\mu$. According to Egorov's theorem for any $\varepsilon > 0$ there is a set $\Lambda_1(\varepsilon)$ such that the sequence $\bar{\gamma}_n$ converges uniformly on $\Lambda_1(\varepsilon)$ and $\mu\Lambda_1(\varepsilon) > 1 - \varepsilon$. As any continuous function on a compact space γ is uniformly continuous, for any $\delta > 0$ there is $\varepsilon(\delta) > 0$ independent of n such that if $y \in B_x^n(\varepsilon)$, then $|\bar{\gamma}_n(x) - \bar{\gamma}_n(y)| < \delta$. Let $\Lambda(q) = \Lambda_1(q) \cap \pi\hat{\Lambda}(q)$, where $\hat{\Lambda}(q)$ is defined as in §3.3. Obviously, $\mu\Lambda(q) \to 1$, as $q \to 0$. On $\Lambda(q)$ the sequence $\bar{\gamma}_n$ converges uniformly, so for big enough n we can assume that $|\bar{\gamma}_n(x) - \int_X \gamma\,d\mu| < \delta$.

$$\begin{aligned}
\int_X e^{n\bar{\gamma}_n(x)} \|Df^n(x)^{\wedge}\|\,dx &> \sum_{x\in(\varepsilon,n,\Lambda(q))\text{-separated}} \int_{B_x^n(\varepsilon)} e^{n\bar{\gamma}_n(z)} \|Df^n(z)^{\wedge}\|\,dz \\
&> \sum_{x\in(\varepsilon,n,\Lambda(q))\text{-separated}} \min_{y\in B_x^n(\varepsilon)} e^{n\bar{\gamma}_n(y)} \int_{B_x^n(\varepsilon)} \|Df^n(z)^{\wedge}\|\,dz \\
&> e^{-\delta n} \sum_{x\in(\varepsilon,n,\Lambda(q))\text{-separated}} e^{n\bar{\gamma}_n(x)} \int_{B_x^n(\varepsilon)} \|Df^n(z)^{\wedge}\|\,dz \\
&> s(n, \varepsilon, \mu\Lambda(q)) e^{\int \gamma\,d\mu} C_5 \varepsilon^{C_6} q^{C_6} e^{-C_7\varepsilon n} e^{-2\delta n}.
\end{aligned}$$

Thus,

$$\liminf_{n\to\infty}\frac{1}{n}\log\int_X e^{\bar\gamma_n(x)}\|Df^n(x)^{\wedge k}\|^p\,dx \quad\geq\quad \underline{r}(\varepsilon,\Lambda(q))+\int_X\gamma\,d\mu-2\delta-C_7\varepsilon.$$

Letting δ tend to 0 we obtain

$$\begin{aligned}\liminf_{n\to\infty}\frac{1}{n}\log\int_X e^{\bar\gamma_n(x)}\|Df^n(x)^{\wedge k}\|\,dx &\geq \sup_{\mu\in\mathcal{M}_e}\left(h_\mu+\int_X\gamma\,d\mu\right)\\ &= P(\gamma).\end{aligned}$$

Part 2. Let $\sigma : I^k \to X$ be a C^∞ k-dim curve, $I=[0,1]$. The following lemma is a modification of [**15**].

LEMMA 4.1. ([**15**]) *Let $\|Df(x)\|<N$ for all $x\in X$. There is $\varepsilon_0>0$ such that for any $\varepsilon<\varepsilon_0$, if the mapping σ has the absolute value of the all derivatives up to the lth order bounded by ε, then*

$$\sup_{x\in X}\mathrm{Vol}_k\, f^n(B_x^n(\varepsilon)\cap\sigma I^k)<\kappa^n,$$

where $\kappa=\alpha(k,l,\dim X)\,(\log N)^{\beta(k,l,\dim X)}\,N^{k/l}$.

LEMMA 4.2. *For $\varepsilon<\varepsilon_0$,*

$$\int_{B_x^n(\varepsilon)}\|Df_x^{n\,\wedge k}\|\,dx<C_8\kappa^n.$$

Proof. Fix $0\leq i_1<\cdots<i_k\leq\dim X$ and $0\leq j_1<\cdots<j_k\leq\dim X$. Let $m^n(x)$ be a minor of Df_x^n such that

$$m^n(x)=\det\begin{pmatrix}\dfrac{\partial f_{i_1}^n}{\partial x_{j_1}}(x) & \cdots & \dfrac{\partial f_{i_1}^n}{\partial x_{j_k}}(x)\\ \vdots & \ddots & \vdots\\ \dfrac{\partial f_{i_k}^n}{\partial x_{j_1}}(x) & \cdots & \dfrac{\partial f_{i_k}^n}{\partial x_{j_k}}(x)\end{pmatrix}.$$

$\int_{B_x^n(\varepsilon)\cap(x_{j_1},\ldots,x_{j_k})\text{-plane}}|m^n(x)|\,dx_{j_1}\cdots dx_{j_k}$ is equal to the k-dim volume with multiplicity of the projection onto the $(x_{i_1},\ldots,x_{i_k})$-plane of the set $f^n(B_x^n(\varepsilon)\cap(x_{j_1},\ldots,x_{j_k})\text{-plane})$. Obviously, the curve $\sigma:(x_{j_1},\ldots,x_{j_k})\mapsto(x_1,\ldots,x_{\dim X})$ satisfies the condition of Lemma 4.1 with arbitrarily large l. Hence, by Lemma 4.1 this volume is less than κ^n. Applying Fubini's theorem we obtain

$$\int_{B_x^n}|m^n(x)|\,dx<\varepsilon^{\dim X-k}\kappa^n<\kappa^n.$$

By Lemma 2.5 the maximum of the k-order minors bounds the k-dim norm, so we are done. □

From now on we follow [**15**]. Set

$$\begin{aligned}t(n,\varepsilon,k,f) &= \sup_{x\in X}\int_{B_x^n(\varepsilon)}\|Df_x^{n\,\wedge k}\|\,dx\\ t(\varepsilon,k,f) &= \limsup_{n\to\infty}\frac{1}{n}\log t(n,\varepsilon,k,f).\end{aligned}$$

LEMMA 4.3. *For* $\varepsilon < \varepsilon_0$,

$$t(\varepsilon, k, f) \leq \log \alpha + \beta \log\log N + \frac{k}{l} \log N.$$

Given $\delta > 0$, define $q(\delta)$ as the minimal integer satisfying

$$\frac{1}{q} (\log \alpha + \beta \log\log N) + \frac{\log q}{q} \beta < \delta.$$

Let $\varepsilon_0(\delta)$ be defined as ε_0 above but for the map $f^{q(\delta)}$ instead of f.

LEMMA 4.4. *For* $\varepsilon < \varepsilon_0(\delta)$,

$$t(\varepsilon, k, f) \leq \delta + \frac{k}{l} \log N.$$

Proof. Clearly, $\max_{x \in X} \|Df_x^{q(\delta)}\| \leq N^{q(\delta)}$. Apply Lemma 4.3 to the map $f^{q(\delta)}$:

$$t(\varepsilon, k, f^q) < \log \alpha + \beta(\log q + \log\log N) + q \frac{k}{l} \log N.$$

It is easy to see that $B^n_{x, f^q}(\varepsilon) \supset B^{qn}_{x, f}(\varepsilon)$, hence $t(n, \varepsilon, k, f^q) \geq t(qn, \varepsilon, k, f)$ and $t(\varepsilon, k, f^q) \geq q t(\varepsilon, k, f)$, so

$$t(\varepsilon, k, f) < \delta + \frac{k}{l} \log N. \qquad \square$$

LEMMA 4.5.

$$\lim_{\varepsilon \to 0} t(\varepsilon, k, f) = 0.$$

Proof. Taking $\delta \to 0$ we get $\lim_{\varepsilon \to 0} t(\varepsilon, k, f) \leq (k/l) \log N$. l is arbitrarily large, so $\lim_{\varepsilon \to 0} t(\varepsilon, k, f) = 0$. $\square$

Now we can finish the proof of Theorem 4.1.

$$\begin{aligned}
\int_X e^{n\bar{\gamma}_n(x)} \|Df_x^{n\,\wedge k}\|\, dx &\leq \sum_{\substack{x \in (n, \varepsilon/2)\text{-separated} \\ \text{and } (n, \varepsilon)\text{-spanning set}}} \int_{B^n_x(\varepsilon)} e^{n\bar{\gamma}_n(y)} \|Df_y^{n\,\wedge k}\|\, dy \\
&\leq \sum_{\substack{x \in (n, \varepsilon/2)\text{-separated} \\ \text{and } (n, \varepsilon)\text{-spanning set}}} \max_{y \in B^n_x(x)} e^{n\bar{\gamma}_n(y)} \int_{B^n_x(\varepsilon)} \|Df_y^{n\,\wedge k}\|\, dy \\
&\leq e^{\delta(\varepsilon)n} t(n, \varepsilon, k, f) \sum_{\substack{x \in (n, \varepsilon/2)\text{-separated} \\ \text{and } (n, \varepsilon)\text{-spanning set}}} e^{n\bar{\gamma}_n(x)} \\
&\leq e^{\delta(\varepsilon)n} t(n, \varepsilon, k, f) r(n, \varepsilon/2, \gamma, f),
\end{aligned}$$

where the function $\delta(\varepsilon)$ has the property that if $y \in B^n_x(\varepsilon)$, then $|\bar{\gamma}_n(x) - \bar{\gamma}_n(y)| < \delta(\varepsilon)$. The discussion in Part 1 of the proof shows that $\lim_{\varepsilon \to 0} \delta(\varepsilon) = 0$.

$$\limsup_{n \to \infty} \frac{1}{n} \int_X e^{n\bar{\gamma}_n(x)} \|Df_x^{n\,\wedge k}\|\, dx \leq \delta(\varepsilon) + t(\varepsilon, k, f) + r(\varepsilon/2, \gamma, f).$$

Letting $\varepsilon \to 0$ we obtain

$$\limsup_{n\to\infty} \frac{1}{n} \int_X e^{n\bar{\gamma}_n(x)} \| Df_x^{n\,\wedge} \| \, dx = \max_k \limsup_{n\to\infty} \frac{1}{n} \int_X e^{n\bar{\gamma}_n(x)} \| Df_x^{n\,\wedge k} \| \, dx \le P(\gamma). \quad \square$$

Proof of Theorem 3.3.

$$\begin{aligned} \int_X \| Df_x^{n\,\wedge k} \|^p \, dx &\le \int_X \| Df_x^{n\,\wedge} \|^p \, dx \\ &\le \int_X \| Df_x^{n\,\wedge} \| \prod_{i=0}^{n} \| Df_x^{i\,\wedge} \|^{p-1} \, dx. \end{aligned}$$

Now, using Theorem 4.1, it is easy to finish the proof. $\square$

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Appendixes

A.1. *Construction of Lyapunov charts.* We include here an outline of the construction of Lyapunov charts for non-invertible maps where it is different from usual. The case of diffeomorphisms can be found in [**10**] or [**6**]. The construction of the inverse limit function was used in [**8**] for similar purposes.

By the multiplicative ergodic theorem [**9**] we know that for any invariant ergodic measure μ there is an invariant set of full measure Γ_1 such that for $x \in \Gamma_1$,

$$\lim_{n\to+\infty} (Df_x^n Df_x^{n*})^{1/2n} = \Lambda_x$$

exists, the eigenvalues $e^{\lambda_1}, \ldots, e^{\lambda_r}$ of Λ_x are independent of x and the splitting of the tangent space into the eigenspaces $T_x X = E_1(x) \oplus \cdots \oplus E_r(x)$ is measurable. Assume that $\lambda_r = -\infty$. Let $U(x)$ be the direct sum of the eigenspaces of Λ_x with non-zero eigenvalues, $U(x) = E_1(x) \oplus \cdots \oplus E_{r-1}(x)$ and let Q_x be the restriction of Df_x on $U(x)$. Since $U(x)$ is measurable, Q_x is also measurable. It is easy to show that

$$\int_X \log |\det Q_x| \, d\mu = \sum_{i=1}^{\dim U} \lambda'_i,$$

and as a consequence $\log |\det Q| \in L_1(X, \mu)$. Hence, $\log \| Q^{-1} \| \in L_1(X, \mu)$. Set $\hat{\Gamma}_1 = \pi^{-1}\Gamma_1$, $\hat{U}(\hat{x}) = U(x)$ and for $\hat{x} \in \hat{\Gamma}_1$ set $\hat{Q}_{\hat{x}} = Q_{\pi\hat{x}}$. Obviously, $\log \| \hat{Q} \| \in L_1(\hat{X}, \hat{\mu})$ and $\log \| \hat{Q}^{-1} \| \in L_1(\hat{X}, \hat{\mu})$. Now we can apply the invertible case of the multiplicative ergodic theorem (see, for instance, [**13**, Theorem 3.1]). There is an invariant set $\hat{\Gamma}$ of full measure and an invariant splitting $\hat{U}(\hat{x}) = \hat{E}_1(\hat{x}) \oplus \cdots \oplus \hat{E}_{r-1}(\hat{x})$, $\hat{x} \in \hat{\Gamma}$, such that for $\hat{v} \in \hat{E}_k(\hat{x})$,

$$\lim_{n\to\pm\infty} \frac{1}{n} \log \| \hat{Q}_{\hat{x}}^n \hat{v} \| = \lambda_k,$$

where

$$\hat{Q}_{\hat{x}}^n = \hat{Q}_{\hat{f}^{n-1}\hat{x}} \cdots \hat{Q}_{\hat{x}} \quad \text{and} \quad \hat{Q}_{\hat{x}}^{-n} = \hat{Q}^{-1}_{\hat{f}^{-n}\hat{x}} \cdots \hat{Q}^{-1}_{\hat{f}^{-1}\hat{x}},$$

if $n > 0$.

This λ_k is the same number as before because the limit when n tends to $+\infty$ is the same in both cases. Obviously $\pi \hat{E}_i(\hat{x}) = E_i(\pi \hat{x})$.

From this point we can proceed exactly as in [**6**].

A.2. *Proof of Lemma 3.1.* From the definition of the multi-dimensional norm if follows that

$$\mathrm{Vol}(Ax_1, \ldots, Ax_k) \leq \|A^{\wedge k}\| \,\mathrm{Vol}(x_1, \ldots, x_k).$$

It implies $\|(AB)^{\wedge k}\| \leq \|A^{\wedge k}\| \|B^{\wedge k}\|$. Obviously, there is a constant C_2 such that

$$\begin{aligned} \|B_i^{\wedge k}\| &< \|A_i^{\wedge k}\| + C_2\varepsilon \\ &< \|A_i^{\wedge k}\| \left(1 + \frac{C_2}{\|A_i^{\wedge k}\|}\varepsilon\right) \\ &< \|A_i^{\wedge k}\| e^{C_3\varepsilon}. \end{aligned}$$

We used the inequality $1+x \leq e^x$. Since $\|A_i^{\wedge k}\| \leq e^{k\varepsilon} e^{\sum_{j=1}^k \lambda'_j}$, $\|B_i^{\wedge k}\| \leq e^{C_4\varepsilon} e^{\sum_{j=1}^k \lambda'_j}$ and $\|(B_n \cdots B_1)^{\wedge k}\| \leq \|B_n^{\wedge k}\| \cdots \|B_1^{\wedge k}\| \leq e^{C_4\varepsilon n} e^{n\sum_{j=1}^k \lambda'_j}$. Thus we have proved the upper bound for $\|(B_n \cdots B_1)^{\wedge k}\|$.

To prove the lower bound first assume that $k = 1$. We can take a cone K around the space $\mathbb{R}^{\dim E_1}$ such that it maps to itself under A_i. Obviously, if we perturb A_i a bit then this cone remains invariant and any vector in it will grow with the rate $\sim \lambda_1$. This proves the lemma in the case $k = 1$. If $k > 1$, we can consider maps $\underbrace{A_i \wedge \cdots \wedge A_i}_{k}$ and apply the same arguments as above.

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